# RANK IS UNCHANGED BY EXTENDING THE GROUND DOMAIN

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Allowed standard prerequisites (editorial).*

- Rank of a module over a domain is the dimension of M tensor K over the fraction field K. definition environment at source lines 5015-5019 of the same chapter, immediately before the counted statement. Supplies the two quantities that are being compared.
- Dimension of a vector space is invariant under extension of the ground field. used at proof lines 5079-5080, the standard statement for field extensions and not a ring-theoretic fact. Carries the dimension of M tensor K over K to the dimension of its base change over the fraction field of R'.
- Base change is compatible with tensoring up to the fraction field for an extension of domains. used implicitly at proof lines 5078-5080 when the rank over R' is identified as the base change of the rank over R. Provides the identification of the two vector spaces to which the invariance of dimension is applied.

*Source reference.* Original lemma 0H9Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21543–21543. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

**Lemma 0.2.** *Let $R \subset R'$ be an extension of domains. If $M$ is an $R$-module, then the rank of $M$ over $R$ is equal to the rank of $M \otimes_R R'$ over $R'$.*

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* This is true because the dimension of a vector space is invariant under extension of ground field. □